CHE 565 -- Homework 2

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Problem 1

Setup

$$\begin{aligned} \text{Option A:} \quad & C_{0,A} = \$3{,}800{,}000 \\ & FV_A = \$1{,}100{,}000/\text{yr} \\ \text{Option B:} \quad & C_{0,B} = \$5{,}000{,}000 \\ & FV_B = \$1{,}410{,}000/\text{yr} \end{aligned}$$

PV = present value, FV = future value (annual cash flow), r = yearly interest rate, n = number of years

n= 10 years, r = 0.10

a.) Compute Net Present Value for Options A and B

$$\text{Present Value formula: PV} = \frac{F \cdot (r+1)^{-n} \cdot ((r+1)^n - 1)}{r}$$

$$\text{Net Present Value formula: NPV} = C_0 + \frac{F \cdot (r+1)^{-n} \cdot ((r+1)^n - 1)}{r}$$

$$\text{Annual payment formula: P} = \frac{C_0 \cdot r \cdot (r+1)^n}{(r+1)^n - 1}$$

$$\text{Annual payment for option A: P}_A = \$492,117.38/\text{year}$$

$$\text{Annual payment for option B: P}_B = \$647,522.87/\text{year}$$

$$\text{NPV}_A = \$2,959,023.82, \quad \text{NPV}_B = \$3,663,839.62$$

NPV is higher for option B at 10% yearly interest, so B is preferred under these assumptions.

b.) Annual Payment for 10 Year Lifetime, No Salvage Value, 5% Interest Rate

P = annual payment, C0 = initial cost, r = yearly interest rate, n = number of years

$$\text{Annual payment for option A: P}_A = \$492,117.38/\text{year}, \quad \text{Annual payment for option B: P}_B = \$647,522.87/\text{year}$$

Problem 2

Setup

\$0.94/lbs is the price of powdered detergent.

0.1 % of 4,000,000 lbs of detergent per year is carried through the exhaust

Adding a second cyclone separator costs \$10,000, with \$300/year in maintenance costs will reduce carryover to 0.0%, with an 8% yearly interest rate.

Cost of carryover detergent: \$3,760.00/year
Cost of second separator: \$10,000.00 initial + \$600.00/year maintenance = \$10,600.00 total

Lost detergent value: \$3,760.00/year

a.) Find the additional yearly income

Gross additional income (recovered detergent): \$3,760.00/year

Net additional income after maintenance: \$3,460.00/year

b.) Find the payback period

$$\text{The NPV equation for the payback period is:} -10000 + 43250.0 \cdot 1.08^{-n} \cdot (1.08^{n} - 1)$$

Discounted payback period (NPV = 0): 3.42 years

Problem 3

Setup

Determine if each function is convex

$$\begin{aligned}
\text{A} &: f(x) = (x_1 - x_2)^2 + x_2^2 \\
\text{B} &: f(x) = x_1^2 + x_2^2 + x_3^2 \\
\text{C} &: f(x) = e^{x_1} + e^{x_2}
\end{aligned}$$

Results

Function	Hessian	Eigenvalues	Convex?	Reason
$x_2^2 + (x_1 - x_2)^2$	$\begin{bmatrix} 2 & -2 \\ -2 & 4 \end{bmatrix}$	$\begin{bmatrix} 3 - \sqrt{5} \\ \sqrt{5} + 3 \end{bmatrix}$	Yes	Eigenvalues are positive
$x_1^2 + x_2^2 + x_3^2$	$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$	$\begin{bmatrix} 2 \end{bmatrix}$	Yes	Eigenvalues are positive
$e^{x_1} + e^{x_2}$	$\begin{bmatrix} e^{x_1} & 0 \\ 0 & e^{x_2} \end{bmatrix}$	$\begin{bmatrix} e^{x_1} \\ e^{x_2} \end{bmatrix}$	Yes	Eigenvalues are non-negative

Problem 4

Is the region defined by the constraints convex?

The feasible region is defined by the following constraints: $x_2 \geq 0 \wedge x_2 \geq 1 - x_1 \wedge x_1 \leq 2 \wedge x_2 \leq \frac{x_1}{2} + 1$

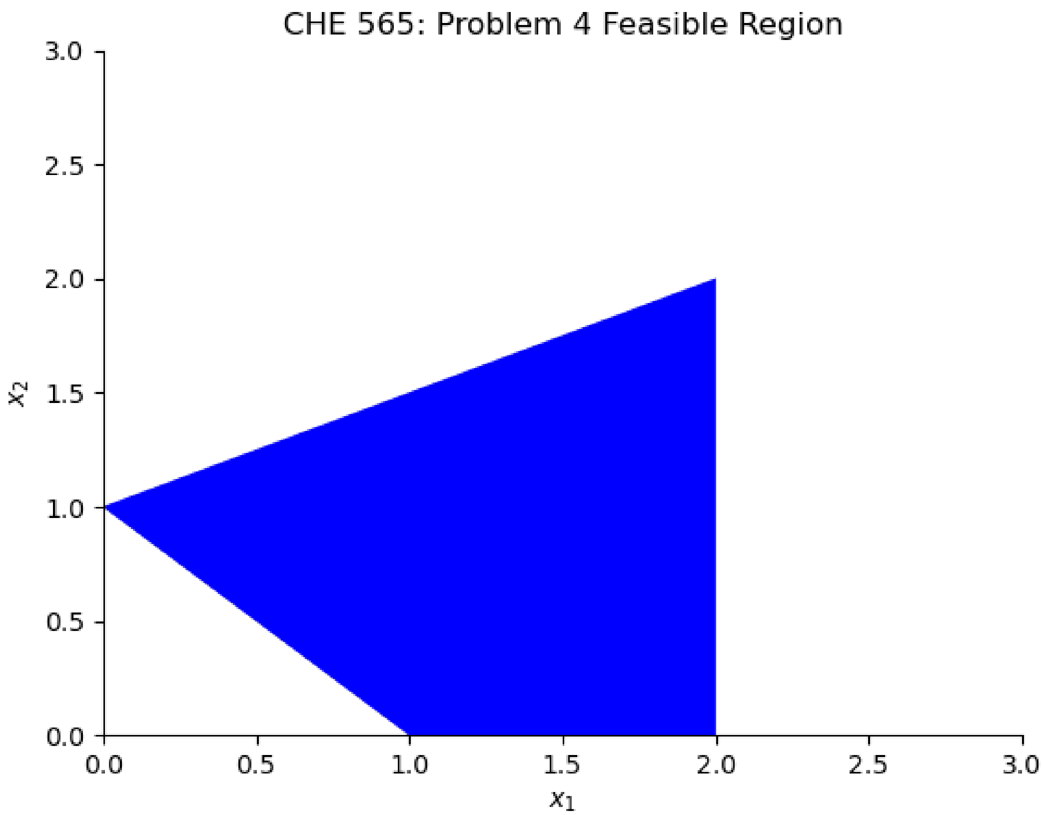

Figure 1: Feasible Region

The feasible region is convex because it is defined by a set of linear inequalities, which form a convex set.

Problem 5

Setup

Maximize the objective function:

$$f(x) = -\frac{x^6}{6} + \frac{4x^5}{5} - \frac{x^4}{4} - \frac{10x^3}{3} + 2x^2 + 8x + 1$$

Given that:

$$\frac{df}{dx} = (2 - x)^3 (x + 1)^2$$

$$\frac{d^2 f}{dx^2} = (2-x)^3 (2x+2) - 3(2-x)^2 (x+1)^2$$

a.) Analytical Solution

Expanded form:

$$\frac{df}{dx} = -x^5 + 4x^4 - x^3 - 10x^2 + 4x + 8$$
$$\frac{d^2 f}{dx^2} = -5x^4 + 16x^3 - 3x^2 - 20x + 4$$

The roots to df/dx are x* = -1, 2

The corresponding function values are f(x*) = -2.88, 9.27

The Hessian evaluated at the critical points are $\frac{d^2 f}{dx^2}|_{x^*}$= 0, 0, inconclusive 2nd derivative test

The maximum value of the function is f(x*) = 9.27 at x* = 2

b.) Excel Solution

The Excel data was used to perform root-finding using the Newton-Raphson method:

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}$$

Iteration	x_n	f(x_n)	f'(x_n)	f"(x_n)	x_n+1
0.0000	5.0000	-586.0833	-972.0000	-1296.0000	4.2500
1.0000	4.2500	-139.2208	-313.9541	-538.2070	3.6667
2.0000	3.6667	-27.0988	-100.8230	-224.6914	3.2179
3.0000	3.2179	0.5555	-32.1432	-94.4150	2.8775
4.0000	2.8775	7.2322	-10.1590	-39.9713	2.6233
5.0000	2.6233	8.8043	-3.1799	-17.0590	2.4369
6.0000	2.4369	9.1644	-0.9854	-7.3392	2.3027
7.0000	2.3027	9.2446	-0.3025	-3.1810	2.2076
8.0000	2.2076	9.2620	-0.0920	-1.3876	2.1413
9.0000	2.1413	9.2657	-0.0278	-0.6084	2.0955
10.0000	2.0955	9.2665	-0.0084	-0.2678	2.0643
11.0000	2.0643	9.2666	-0.0025	-0.1182	2.0432
12.0000	2.0432	9.2667	-0.0007	-0.0523	2.0289
13.0000	2.0289	9.2667	-0.0002	-0.0232	2.0193
14.0000	2.0193	9.2667	-0.0001	-0.0103	2.0129
15.0000	2.0129	9.2667	-0.0000	-0.0046	2.0086
16.0000	2.0086	9.2667	-0.0000	-0.0020	2.0058
17.0000	2.0058	9.2667	-0.0000	-0.0009	2.0038
18.0000	2.0038	9.2667	-0.0000	-0.0004	2.0026
19.0000	2.0026	9.2667	-0.0000	-0.0002	2.0017
20.0000	2.0017	9.2667	-0.0000	-0.0001	2.0011

Table 1: Optimization Results from Excel starting at x0=5

Iteration	x_n	f(x_n)	f'(x_n)	f"(x_n)	x_n+1
0.0000	-5.0000	-4832.7500	5488.0000	-5096.0000	-3.9231
1.0000	-3.9231	-1208.5512	1775.5079	-2114.1044	-3.0832
2.0000	-3.0832	-295.6352	570.0315	-883.6738	-2.4382
3.0000	-2.4382	-71.0801	180.8132	-373.6709	-1.9543
4.0000	-1.9543	-17.8527	56.3066	-160.7261	-1.6040
5.0000	-1.6040	-5.9169	17.0747	-70.7559	-1.3626
6.0000	-1.3626	-3.4409	5.0003	-32.0380	-1.2066
7.0000	-1.2066	-2.9756	1.4068	-14.9373	-1.1124
8.0000	-1.1124	-2.8972	0.3808	-7.1437	-1.0591
9.0000	-1.0591	-2.8853	0.0999	-3.4805	-1.0304
10.0000	-1.0304	-2.8836	0.0257	-1.7158	-1.0154
11.0000	-1.0154	-2.8834	0.0065	-0.8515	-1.0078
12.0000	-1.0078	-2.8833	0.0016	-0.4242	-1.0039
13.0000	-1.0039	-2.8833	0.0004	-0.2117	-1.0020
14.0000	-1.0020	-2.8833	0.0001	-0.1057	-1.0010
15.0000	-1.0010	-2.8833	0.0000	-0.0528	-1.0005
16.0000	-1.0005	-2.8833	0.0000	-0.0264	-1.0002
17.0000	-1.0002	-2.8833	0.0000	-0.0132	-1.0001
18.0000	-1.0001	-2.8833	0.0000	-0.0066	-1.0001
19.0000	-1.0001	-2.8833	0.0000	-0.0033	-1.0000
20.0000	-1.0000	-2.8833	0.0000	-0.0017	-1.0000

Table 2: Optimization Results from Excel starting at x0=-5

x0	x*	f(x*)
5.0000	2.0011	9.2667
-5.0000	-1.0000	-2.8833

Table 3: Summary of Optimization Results from Excel

c.) Numerical solution using Nelder-Mead Simplex algorithm which is the same as fminsearch in matlab

Using Nelder-Mead Simplex algorithm with initial guesses x0 = 5:

One of the roots of df/dx are found to be x* = 1.9999

The corresponding optimum function values are f(x*) = 9.2667

Actual maximum of the function:

$$
\begin{aligned}
x^* &= 1.9999 \\
f(x^*) &= 9.2667 \\
\text{iterations} &= 20 \\
\text{function calls} &= 41
\end{aligned}
$$

d.) Numerical solution using Newton Conjugate Gradient Method which is the same as fmincon in matlab except required to give jacobian and hessian information

Using the Newton Conjugate Gradient method with initial guesses x0 = 5:

One of the roots of df/dx are found to be x* = 2.0038

The corresponding optimum function values are f(x*) = -9.2667

$$\begin{aligned} x^* &= 2.0038 \\ f(x^*) &= -9.2667 \\ \text{iterations} &= 19 \\ \text{function calls} &= 19 \end{aligned}$$

Using the Newton Conjugate Gradient method with initial guesses x0 = -5:

One of the roots of df/dx are found to be x* = -1.0001

The corresponding optimum function values are f(x*) = 2.8833

$$\begin{aligned} x^* &= -1.0001 \\ f(x^*) &= 2.8833 \\ \text{iterations} &= 20 \\ \text{function calls} &= 20 \end{aligned}$$

e.) Compare results from all methods

Method	Initial Guess	x*	f(x*)	Iterations	Function Calls
Nelder-Mead	5	1.9999	-9.2667	20	41
Excel	-5	2.0011	9.2667	21	21
Excel	5	-1.0000	-2.8833	21	21
Newton-CG	5	2.0038	-9.2667	19	19
Newton-CG	-5	-1.0001	2.8833	20	20

All methods found the same maximum value of the function at x* = 2 and f(x*) = 9.267, which is consistent with the analytical solution. The Nelder-Mead method required 20 iterations and 41 function calls, the Newton-CG method converged in 20 iterations, the Excel solution also converged to the same optimum value, demonstrating consistency across methods. However, Newton-CG and the excel solution are significantly more influenced by the initial guess, and can converge to a local extremum. In the case of this polynomial function, there are two critical points based on the roots of the derivative, one is an optimum at x* = 2 and the other is a saddle point at x* = -1, and the initial guess determines which extremum the algorithm converges to. The Nelder-Mead method is less sensitive to the initial guess and was able to find the global maximum regardless of the starting point.